

A New Similarity Space Tailored for Supervised Deep Metric Learning

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We propose a novel deep metric learning method. Differently from many works in this area, we define a novel latent space obtained through an autoencoder. The new space, namely S-space, is divided into different regions describing positions where pairs of objects are similar/dissimilar. We locate makers to identify these regions and estimate the similarities between objects through a kernel-based Cauchy distribution to measure the markers' distance and the new data representation. In our approach, we simultaneously estimate the markers' position in the S-space and represent the objects in the same space. Moreover, we propose a new regularization function to prevent similar markers from collapsing altogether. Our method emphasizes the group property (separability) while preserving instance representativity. We present evidence that our proposal can represent complex spaces, for instance, when groups of similar objects are located in disjoint regions. We compare our proposal to nine different distance metric learning approaches (four of them are based on deep learning) on 28 real-world heterogeneous datasets. According to the four quantitative metrics used, our method overcomes all of the nine strategies from the literature.

CCS Concepts: • **Computing methodologies** → **Learning latent representations**; **Neural networks;**

Additional Key Words and Phrases: Similarity space, deep metric learning, latent feature space, regularization function

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1 INTRODUCTION

A *distance metric* is a function that provides a way to measure how far apart two elements of a set are from each other. Among various works involving machine learning applications, the most commonly used metric is the Euclidean distance [10]. Methods that use Euclidean distance usually consider that all variables' covariance is zero (i.e., there is no correlation among them, but this assumption is hardly found in the real world) [64]. Euclidean distance and cosine similarity are popular for many applications. For instance, the cosine similarity is vastly used for text mining [27]. Even showing its effectiveness in several applications, the cosine similarity assumes equal weight for every dimension, limiting its application [27].

Euclidean and cosine distance are known as data-independent techniques once they are defined without prior knowledge about the data. Learning distances, a.k.a. **Metric Learning (MeL)**, from data is a common attempt to improve machine learning approaches [12, 15, 21, 22, 28, 54, 60]. In modern machine learning research, MeL is a fundamental technique for several different applications, such as sorting [33], classification (e.g., ***k*-Nearest Neighbor (KNN)**), clustering [63], and ranking [53].

MeL aims to estimate distance function parameters based on a given training set. A common approach is to frame MeL as a convex optimization problem [66]. Thus, a distance d can be defined as $d_{\boldsymbol{M}}(x, y) = \sqrt{(x-y)^T \boldsymbol{M}(x-y)}$, in which $\boldsymbol{M}$ is a positive semi-definite matrix. In the case $\boldsymbol{M}$ is the covariance matrix, we have the *Mahalanobis* distance [10]. Classic methods proposed for MeL use $d_{\boldsymbol{M}}$ to search for the best linear space that captures the semantics of the data (e.g., in a classification setting, we search for $\boldsymbol{M}$ that minimizes the miss-classification loss). However, the linear transformation has some limitations, as it cannot model high-order correlations between the original data dimensions [4].

Using MeL, we can define metrics that consider the covariance of attributes. In addition, MeL approaches do not necessarily assume linear relationships, although classical MeL techniques like the Mahalanobis distance [10] assume a linear space. Moreover, MeL does not assume equal weights for every attribute [27]. The assumption that MeL can be treated as a convex optimization problem can also be relaxed using the appropriate model.

Deep learning techniques are currently being used for MeL to tackle the issues mentioned previously [20, 32, 39, 41, 46, 70]. Since these proposals seek to learn a non-linear feature representation, they usually overperform standard techniques in the literature. **Neural Networks (NNs)** are natural candidates and are typically used to learn similarity metrics [7, 17].

The representation of compressed data found by an NN is commonly named *latent feature space*, and the data in this space is latent data, as we can see in Figure 1(a). Our work hypothesizes that the latent feature space captured by NNs can be improved with an auxiliary space. For instance, typical NN-based **Deep Metric Learning (DMeL)** approaches extract a latent space that encodes similar and dissimilar *points*, but not the separability among them. However, this single representation is limited, as it does not capture *pairwise information*.

Unlike the literature, our approach employs NNs, fed by labeled original pairwise data, to find a *latent pairwise space with markers*. This approach is shown in Figure 1(b), as we now detail. In our method, data comes in pairs of vectors $(\mathbf{x}_i, \mathbf{x}_j)$ that are deemed as similar ($\boldsymbol{x}_i$ has the same label as $\boldsymbol{x}_j$) or dissimilar ($\boldsymbol{x}_i$ has the different label as $\boldsymbol{x}_j$). The first part of our architecture is an autoencoder. After encoding the pair of input objects, our major novelty is on converting *data pairs* to a new **Similarity space (S-space)**. A data point $\mathbf{x}_i$ is mapped into $\mathbf{z}_i = f_\Theta(\mathbf{x}_i)$ in the latent space, where Θ are model parameters. The S-space for a pair of points i and j is composed of two novel ideas. First, we represent points as a similarity vector between pairs (i.e., $\mathbf{s}_{ij} = |\mathbf{z}_i - \mathbf{z}_j|$). Second, *and more importantly*, we define markers that act as reference points to similar ($\boldsymbol{\mu}_p^+ \in \mathcal{M}^+$) and dissimilar ($\boldsymbol{\mu}_n^- \in \mathcal{M}^-$) regions. Markers' positions are learned in the optimization process.

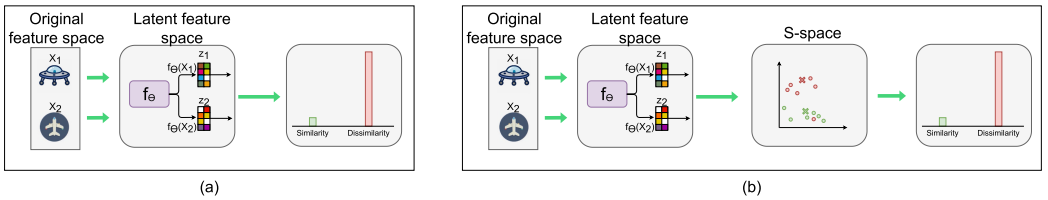


Fig. 1. Comparison between the canonical model of Deep Metric Learning (DMeL) and the model used in this work. (a) Canonical scheme of DMeL. (b) Our scheme of DMeL with S-space.

Our loss function is comprised of three parts. First, an autoencoder loss function takes care of data encoding and decoding. The second loss function captures the sum of distances between similarity vectors ($\mathbf{s}_{ij}$) and markers ($\boldsymbol{\mu}_m \in \mathcal{M}^+ \cup \mathcal{M}^-$); in this work, we used a kernel-based discrete Cauchy distribution to estimate this distance, and we apply a cross-entropy loss function between the input labels and the model output. The last part of our loss function is called a *repulsive regularizer*. It is inversely proportional to the distance of the markers of the same class. This loss function ensures that markers are different (the loss increases as markers become similar), ensuring some diversity level on the marker set. It attempts that markers capture complex similarity regions such as disjoint similarity/dissimilarity regions.

We named our approach **Supervised distance MEtric Learning encoder with simiLarity space (SMELL)**. Our method is herein described as supervised learning, but it can be appropriately extended to unsupervised [30] and semi-supervised learning [69] with some pseudo-label approaches [5, 13]. For example, in pseudo-label [5], the labeled data are categorized via k-means clustering, and patterns' data are sampled to train multimodal. In addition, for unlabeled data, if multimodal reach an agreement on predicting a sample, this sample is labeled; otherwise, keep it unlabeled. Through a wide range of experiments on 28 datasets, we show that SMELL provides gains over the state of the art in all of them. To explain its accuracy, we show evidence supporting the following two hypotheses.

Hypothesis 1. (H1) SMELL groups data points considered similar (in our context, which have the same labels) and dissimilar (different labels) into disjoint regions in S-space.

Hypothesis 2. (H2) SMELL increases the input pairs' separability in the latent feature space for different pairs (similar/dissimilar) types.

Overall, the main contributions of our work are the following:

(1) We present a new data representation space called*S-space* that separates regions where similar/dissimilar objects lie together and help the convergence of the model. We also investigate interpretability and data visualization in this space. S-space can capture complex regions that can model similar points in disjoint regions.
(2) We present a new distance MeL method that simultaneously learns a latent representation of the data and the markers' position in the S-space.
(3) We found evidence that the number of markers is a virtual hyperparameter of the model and does not need to be tuned.
(4) We present a new regularization function to avoid model overfitting called *repulsive regularizer*.

This article is organized as follows. Section 2 presents the related works to DMeL. Section 3 describes some notations and a background review for the good understanding our proposal. Section 4 describes our proposal. Section 5 describes the experimental setup used to analyze the data. Section 6 presents the main results and discussions, and Section 7 presents our conclusion.

2 RELATED WORK

In the distance MeL task, prior research usually assumes that the datasets are represented by an incomplete set of features (i.e., we can never collect all features of an object). This subset of features may not thoroughly inform the semantics of the data space. Thus, the objective is to learn a similarity matrix that encodes how these features should be combined to compute distances best.

One of the first successful cases to solve this problem was learning the linear matrix (Mahalanobis) metric to find a new representation in the feature space [14, 23, 67]. This paradigm requires the decomposition of eigenvalues, an operation that is cubic in the dataset dimensionality (i.e., number of features). This issue severely impacts the training time. In addition, approaches like this are limited to similarity matrices, which encode linear combinations of features.

Other approaches proposed techniques based on information theory to tackle the distance MeL problem [9, 34, 38]. These works start from a reference distribution to train distance functions based on divergences (e.g., Kullback-Leibler or Jeffrey) to obtain reference probability distributions of the data. Through this reference distribution, the authors estimate the similarity. These methods usually suffer from convergence issues [9] when optimizing.

In kernel-based methods for distance MeL, the input data is usually transformed into a higher-dimensional space. The algorithm learns object similarities using the new space obtained from the kernel function [34, 36, 50, 58]. These methods also suffer from a cubic computation cost (on the number of features) or convergence issues, limiting their applicability due to training time.

In the context of deep learning methods, a typical family of deep NN models that learns distance metrics is the Siamese NNs. One of the first works using this approach can be seen in the work of Bromley et al. [2], where the authors propose a model composed of two NNs that share their weights among themselves. This architecture was initially proposed for the signature verification problem.

NNs seek to find non-linear similarities between comparable data examples by extracting a feature vector representing the difference between the data examples. There are several works in the context of NNs developed for different applications [6, 7, 24, 45, 62, 68]. They are easily scalable (do not suffer from the cubic cost as before), as they do not explore eigenvalues decomposition. NNs are typically optimized with functions that consider pairs of inputs, called *pairwise loss functions*, and these proposals tend to find a new representation of the data. Therefore, a similarity function is defined in this new representation (e.g., Euclidean).

More recent works present DMeL with contrastive loss [7, 17] and triplet loss [44]. Even showing promising results, these proposals present some issues, such as slow convergence and poor local optima, making optimizing the model challenging. Contrastive embedding is highly dependent on the quality of the representation of the training data. The training set must contain real-valued precision for pairwise samples. This consideration is typically difficult to satisfy, which is usually not available in practice [57]. For the triplet model, the loss function defines an inequality between positive and negative examples for a given anchor example. These methods suffer from the hard negative problem [37, 61]. Here, some specific negative examples deteriorate the quality of the model, making the training unstable [8]. Hard negative data mining is a proposal to work around this problem. However, the computational cost of searching for these examples becomes high. In addition, it is unclear what defines "good" hard triplets [47].

Recent work, including N-pair loss [48], lifted structure [40], and the multi-similarity loss [59], proposes strategies to capture relationships within a mini-batch selection. Typically, these strategies consider a weight function that associates the pairs of elements in the loss calculation. Nevertheless, these works are based on distance measurements between pairs of similar and dissimilar objects in the space found by the NN. In addition, this characteristic of approximating points

Table 1. Notation Used in This Article

Notation	Description
$\mathcal{X}$	Input data examples set
$\boldsymbol{x}_i$	m-Dimensional single element in $\mathcal{X}$
$\mathcal{Y}$	Label set for set $\mathcal{X}$
y_i	Single element in $\mathcal{Y}$
$\mathcal{Z}$	Latent feature space from $\mathcal{X}$
$\boldsymbol{z}_i$	n-Dimensional single element in $\mathcal{Z}$
f_Θ	Encoder function
Θ	Set of weights for encoder
$f_{\Theta'}$	Decoder function
Θ'	Set of weights for decoder
l	Label function for a element in set $\mathcal{X}$
l'	Label function for a element in set $\mathcal{Z}$
$\mathcal{S}$	*S-space* from $\mathcal{Z}$
$\boldsymbol{s}_{ij}$	n-Dimensional single element in $\mathcal{S}$
$\mathcal{M}$	*Markers set*
μ_i	n-Dimensional single element in $\mathcal{M}$
f^S	Function that maps a pair in $\mathcal{X}$ to an element in $\mathcal{S}$
ψ	Similarity function
Σ	Set of parameters of ψ (Θ, Θ', and $\mathcal{M}$)

considered similar can be found in grouping proposals (e.g., [26, 31, 42]). However, clustering approaches estimate a new representation in the latent feature space. The methods mentioned in this section indicate the feasibility of learning a similarity function from the input data. Some of these methods inspire the present work [2, 31]—for instance, we use NNs to extract the data representation, Contrastive clustering, and the Cauchy kernel distribution to create a similarity metric.

However, we devised a novel DMeL method different from the literature using a new representation space (S-space) obtained through autoencoders. As defined herein, the S-space helps the convergence of the proposal, and thanks to the possibility of having multiple markers to represent similar objects, it models even complex spaces such as non-continuous spaces where similar objects lie in disjoints regions. Therefore, we propose a new S-space that helps the learning of autoencoders. Unlike pairwise loss, our proposal does not require any specific sample selection strategy.

3 BACKGROUND AND NOTATION

In SMELL, we map pairwise input data into a latent space and an S-space. In this section, we provide some technical background about data representation with autoencoders and a mathematical notation essential to the proposed method understanding.

Throughout the article we apply the following notation. We denote vectors by boldface lowercase letters, such as $\boldsymbol{x}$, $\boldsymbol{z}$, and $\boldsymbol{\mu}$; all scalars by lowercase letters, such as m and n; sets of parameters by Greek uppercase letters, such as Θ and Σ; and sets by calligraphic uppercase letters, such as $\mathcal{X}$ and $\mathcal{Z}$. The zero-mean normal distribution will be denoted by $\mathcal{N}(\mu = 0, \sigma)$. Table 1 summarizes this notation.

Let the set $\mathcal{X} = \{\boldsymbol{x}_i\}_{i=1}^{v}$, with $\boldsymbol{x}_i \in \mathbb{R}^m$, be v data examples defined in an m-dimensional feature space. For each $\boldsymbol{x}_i \in \mathcal{X}$, there is an associated label $y_i \in \mathcal{Y} = \{y_i\}_{i=1}^{v}$, where $y_i \in \{1, \ldots, b\}$. In this

way, the pair $(\boldsymbol{x}_i, y_i)$ indicates which of b classes an input $\boldsymbol{x}_i$ belongs to. In a supervised machine learning classification problem, we seek to find a function $l : \mathcal{X} \rightarrow \mathcal{Y}$ that maps an unlabeled example $\boldsymbol{x}_i$ into their respective label y_i. To develop the proposed work, we introduce here some important definitions.

Definition 3.1 (The Latent Feature Space). Consider the set $\mathcal{X}$ as the original feature space and the representation function $f_\Theta : \mathcal{X} \rightarrow \mathcal{Z}$, in which $f_\Theta(\boldsymbol{x}_i) = \boldsymbol{z}_i \implies l(\boldsymbol{x}_i) = l'(\boldsymbol{z}_i) = y_i$ and the function $l' : \mathcal{Z} \rightarrow \mathcal{Y}$, which maps the latent data into their respective labels. We defined the representation space $\mathcal{Z}$ called *latent feature space* from $\mathcal{X}$ as $\mathcal{Z} = \{\boldsymbol{z}_i\}_{i=1}^{v}$, with$\boldsymbol{z}_i \in \mathbb{R}^n$.

An autoencoder is an NN trained to attempt to copy a data input to its output. It can be seen as consisting of two parts: an encoder and a decoder that produces an input-based reconstruction [16].

An encoder is a representation learning algorithm that seeks to find a representation function $f_\Theta : \mathcal{X} \rightarrow \mathcal{Z}$ for a set of weights Θ that maps the set $\mathcal{X}$ to the latent feature space $\mathcal{Z}$. Similarly, the decoder function can be defined as the inverse encoder function $f_{\Theta'}^{-1} : \mathcal{Z} \rightarrow \mathcal{X}$, where Θ' is a set of weights for the decoder. Autoencoders are trained to minimize reconstruction errors (typically, mean squared errors (MSEs)), and their training is performed through *backpropagation* of the error, just like a regular feedforward NN [18].

A Siamese NN model [2, 59] receives a pair of input examples $(\boldsymbol{x}_i, \boldsymbol{x}_j) \in \mathcal{X} \times \mathcal{X}$ and transforms each of them to a latent data $(\boldsymbol{z}_i, \boldsymbol{z}_j) \in \mathcal{Z} \times \mathcal{Z}$ through the encoder f_Θ. In the context of supervised learning, for a data pairwise $(\boldsymbol{x}_i, \boldsymbol{x}_j) \in \mathcal{X} \times \mathcal{X}$, we say they are similar iff $l(\boldsymbol{x}_i) = l(\boldsymbol{x}_j)$. Analogously, they are dissimilar iff $l(\boldsymbol{x}_i) \neq l(\boldsymbol{x}_j)$.

Definition 3.2 (The S-space). The representation space called *S-space* is a space built from the set $\mathcal{X} \times \mathcal{X}$. So, let f^S be the function $f^S : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{S}$, the *S-space* is defined as $\mathcal{S} = \{\boldsymbol{s}_{ij}\}$, with$\boldsymbol{s}_{ij} \in S \subset \mathbb{R}^n$, where $\boldsymbol{s}_{ij}$ represents the similarity vector if $l(\boldsymbol{x}_i) = l(\boldsymbol{x}_j)$, and if $l(\boldsymbol{x}_i) \neq l(\boldsymbol{x}_j)$, then $\boldsymbol{s}_{ij}$ represents the dissimilarity vector.

In this article, we define the map function $f^S : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{S}$ for a pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ by the following element-wise absolute value operation:

$$\boldsymbol{s}_{ij} = f^S(\boldsymbol{x}_i, \boldsymbol{x}_j) = |f_\Theta(\boldsymbol{x}_i) - f_\Theta(\boldsymbol{x}_j)| = |\boldsymbol{z}_i - \boldsymbol{z}_j| = \left(|z_i^1 - z_j^1|, |z_i^2 - z_j^2|, \cdots, |z_i^n - z_j^n|\right). \quad (1)$$

It is worth noting that since $\boldsymbol{s}_{ij}$ is obtained by an element-wise process, it has the same dimension as $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$, where z_i^n is the n-th feature of the i-th data example in a latent space representation $\mathcal{Z}$ (see Definition 3.1). In *S-space*, we define similarity and dissimilarity markers to group $\boldsymbol{s}_{ij}$ vectors into their respective representations. The markers are vectors in the S-space, which have their positions estimated by SMELL—that is, in training the model, we estimate the NN's weights and the markers' position. Thus, the closer $\boldsymbol{s}_{ij}$ is to the similarity/dissimilarity marker, the greater the probability that the $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is similar/dissimilar.

Definition 3.3 (The Markers Set). In *S-space*, we defined the Markers set $\mathcal{M} \subset \mathbb{R}^n$ (same space as $\mathcal{S}$) to improve similarity calculations. We define the set $\mathcal{M}^+$ representing the set of markers responsible for quantifying the similarity between the input pairs. Likewise, markers in set $\mathcal{M}^-$ quantify the dissimilarity. The Markers set is defined as $\mathcal{M} = \mathcal{M}^+ \cup \mathcal{M}^- = \{\boldsymbol{\mu}_i^+\}_{i=1}^{k} \cup \{\boldsymbol{\mu}_j^-\}_{j=k+1}^{w}$.

Therefore, in this work, we seek to calculate the similarity function $\psi_\Sigma : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$. The parameters of ψ are defined by the set $\Sigma = \{\Theta, \Theta', \mathcal{M}\}$, respectively the weights of the encoder, decoder, and the Markers set in S-space. SMELL relies upon simultaneously learning all elements of Σ. More details about the proposed method are described in Section 4.

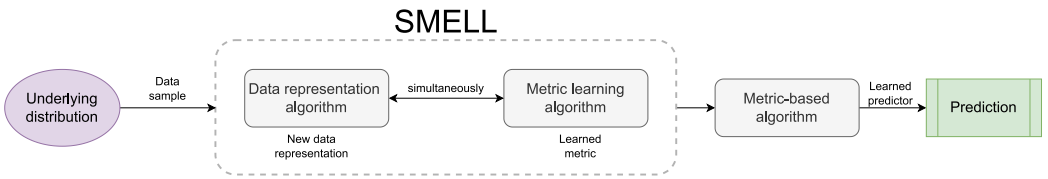


Fig. 2. Simple black box schematic for our proposal.

4 SUPERVISED DISTANCE METRIC LEARNING ENCODER WITH SIMILARITY SPACE

Our proposal, SMELL, simultaneously optimizes a latent data representation (using a DMeL model) and a similarity function that indicates the similarity of two objects in the learned data S-space. This technique can be useful for various applications, such as feeding a predictor (e.g., a classifier) with a new metric learned from the data. This section details our proposal. Figure 2 shows a simple schematic for our proposal.

4.1 MeL Algorithm

There are several ways to find a similarity metric ψ_Σ [1, 56]. In this article, we propose ψ_Σ being estimated from the latent representation obtained by the encoder $f_\Theta : \mathcal{X} \rightarrow \mathcal{Z}$.

4.2 S-space

As seen in Definition 3.2, we define a new representation space, S-space $\mathcal{S}$, which quantifies the similarity between pairs of objects. In Equation (1), we propose a map function $f^S : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{S}$ for a data pairwise $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ as being an element-wise absolute value operation representing the pairwise difference between the pair of data. Note, in Equation (1), that $\boldsymbol{s}_{ij} \in \mathbb{R}^n$ (same dimension as latent representation space).

Regarding the pairwise labeling, we have two options for a given pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$: similar or dissimilar.

Thus, we define the *Markers set* $\mathcal{M}$ so that each marker of $\mathcal{M}^+$ or $\mathcal{M}^-$ represents one of these possibilities (see Definition 3.3). The closer the vector $\boldsymbol{s}_{ij}$ is to a marker $\boldsymbol{\mu}^+ \in \mathcal{M}^+$ or $\boldsymbol{\mu}^- \in \mathcal{M}^-$, the greater the probability that the elements of the pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ are similar or dissimilar to each other, respectively. Then, we have, in this case, k similarity markers and $w - k$ dissimilarity markers for $\mathcal{M}$, and $\mathcal{M}^+ \cap \mathcal{M}^- = \emptyset$.

Inspired by other works [25, 31, 65], we use Student's t-distribution with 1 degree of freedom (Cauchy distribution) as a kernel to measure the similarity between $\boldsymbol{s}_{ij}$ and a specific marker $\boldsymbol{\mu}_m \in \mathcal{M}$, as

$$q_{ij}^m = \frac{\left(1 + ||\boldsymbol{s}_{ij} - \boldsymbol{\mu}_m||_2^2\right)^{-1}}{\sum_{\mu_{m'} \in \mathcal{M}} \left(1 + ||\boldsymbol{s}_{ij} - \boldsymbol{\mu}_{m'}||_2^2\right)^{-1}}, \tag{2}$$

where $q_{ij}^m \in \mathbb{R}$ is the similarity/dissimilarity of $\boldsymbol{s}_{ij}$ in relation to the markers μ_m (it is normalized by the sum of all markers in $\mathcal{M}$). So, we calculate $q_{ij}^+ = \sum_p q_{ij}^p$ for all $\boldsymbol{\mu}_p \in \mathcal{M}^+$ and $q_{ij}^- = \sum_n q_{ij}^n$ for all $\boldsymbol{\mu}_n \in \mathcal{M}^-$. In other words, q_{ij}^+ is the probability of $\boldsymbol{x}_i$ having the same label as $\boldsymbol{x}_j$ and q_{ij}^- is the probability of $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ having different labels. Since $\mathcal{M}^+$ and $\mathcal{M}^-$ are two disjoint sets, we have $q_{ij}^+ + q_{ij}^- = 1$.

It is worth noting that we use a different version of the DMeL canonical model. Thus, we use the representation of the difference vector $\boldsymbol{s}_{ij}$ defined in S-space. In Section 6.4, we show more details about this choice. Finally, as shown in Equation (2), we conduct an iterative sum after obtaining data embeddings (in S-space). So, our proposal adds the time complexity $O(w)$, where d is the

latent feature space dimension (typically $d < 1024$, and w is the number of markers (typically $w < 8$). Furthermore, $O(dw) \ll O(N)$, where $O(N)$ is the complexity of the NN training (encoder function).

4.3 Loss Function

SMELL relies on simultaneously learning a latent representation of the data (with parameters Θ and Θ' for the encoder and decoder functions, respectively) and the positioning of the markers of the set $\mathcal{M}$ in S-space. Therefore, finding the parameters $\Sigma = \{\Theta, \Theta', \mathcal{M}\}$ of the function $\psi_\Sigma(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is defined as an optimization problem. Our loss function is composed of the weighted sum of three loss functions. Let our loss function be j, and we estimate the optimal parameters set Σ^* with cross-entropy loss H_c (first loss function). We define a regularization functions R_r (second loss function) and R_d (third loss function) to avoid overfitting in the training process. In training, cross entropy is applied between the output of SMELL and the object's classes.

As the cross-entropy loss focuses only on improving the object's separability in the S-space, a loss function composed by the cross-entropy loss alone may lead the S-space to distort the latent feature space, weakening the representativeness of embedded features. As shown in the work of Peng et al. [43], autoencoders can preserve the local structure of underlying data distribution. So, similarly to the work of Dizaji et al. [13], R_r regards the autoencoder's reconstruction error. In our proposal, for all training pairs $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ and for all reconstructed pairs $(\boldsymbol{x}'_i, \boldsymbol{x}'_j)$, we have $R_r(\boldsymbol{x}_i, \boldsymbol{x}_j) = r_r[||\boldsymbol{x}_i - \boldsymbol{x}'_i||_2^2 + ||\boldsymbol{x}_j - \boldsymbol{x}'_j||_2^2]$, where r_r is a constant to calibrate the loss reconstruction function, and N is the number of pairs in training the dataset. Under this condition, fine-tuning SMELL will not cause corruption in latent feature space.

When we use more than one marker as reference points to the similarity/dissimilarity regions, markers of the same set $\mathcal{M}^+$ (or $\mathcal{M}^-$) tend to group together, hindering the efficiency of our method. In this context, we propose a new regularization term R_d that we called *repulsive regularizer* to avoid this undesirable behavior. It is defined as

$$R_d^+ = \frac{1}{c^+}\left[\sum_{\mu_i \in \boldsymbol{M}^+}\sum_{\mu_j \in \boldsymbol{M}^+}\frac{1}{||\mu_i - \mu_j||_2^2 + \epsilon}\right], \tag{3}$$

where $\boldsymbol{\mu}_i \neq \boldsymbol{\mu}_j$ and c^+ is a constant value defined as $c^+ = \binom{k}{2}$, in which k is the number of elements in $\mathcal{M}^+$ (see Definition 3.3). R_d is inversely proportional to the square distance of the markers. To avoid ill-formed problems, we added a corrective term ϵ to the denominator that prevents division by 0. We conducted a manual investigation with a grid search, and we adopted $\epsilon = 10^{-1}$ for our experiments. In the same way, we define R_d^-, and with that, we have $R_d = r_d[R_d^+ + R_d^-]$, with a constant value r_d for calibration. Note that $R_d^+ = 0$ if we have a single positive marker $k = 1$. In the same way, if we have a single negative marker, $R_d^- = 0$ if $w - k = 1$.

Let $\mathcal{Q} = \{q_{ij}\}$, the SMELL output, be the set that contains the pairs $q_{ij} = (q_{ij}^+, q_{ij}^-)$ corresponding to the probability of the elements of a pairwise input $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ be similar or dissimilar, respectively. The optimal hyperparameters set can be defined as $\Sigma^* = \arg\min_\Sigma j$, where

$$j = R_d + \sum_{i \in \mathcal{X}}\sum_{j \in \mathcal{X}} J(\boldsymbol{x}_i, \boldsymbol{x}_j) = R_d + \sum_{i \in \mathcal{X}}\sum_{j \in \mathcal{X}}\left[r_{HC} H_c(\boldsymbol{u}_{ij}, \boldsymbol{q}_{ij}) + R_r(\boldsymbol{x}_i, \boldsymbol{x}_j)\right], \tag{4}$$

where r_{HC} is a constant for calibration, $H_c(\boldsymbol{u}_{ij}, \boldsymbol{q}_{ij})$ is the binary cross-entropy loss, and $\boldsymbol{u}_{ij} \in \mathcal{U}$ is defined as $\boldsymbol{u}_{ij} = (1, 0)$ if i has same label as j and $\boldsymbol{u}_{ij} = (0, 1)$ otherwise.

SMELL simultaneously learns all parameters in the set Σ^*. The representation found in S-space aims at grouping the elements $\boldsymbol{s}_{ij}$ around their respective markers, as defined in loss function j (Equation (4)). According to the similarity function, the closer the $\boldsymbol{s}_{ij}$ is to the similarity/dissimilarity marker, the greater the probability that the S-vector is similar/dissimilar.

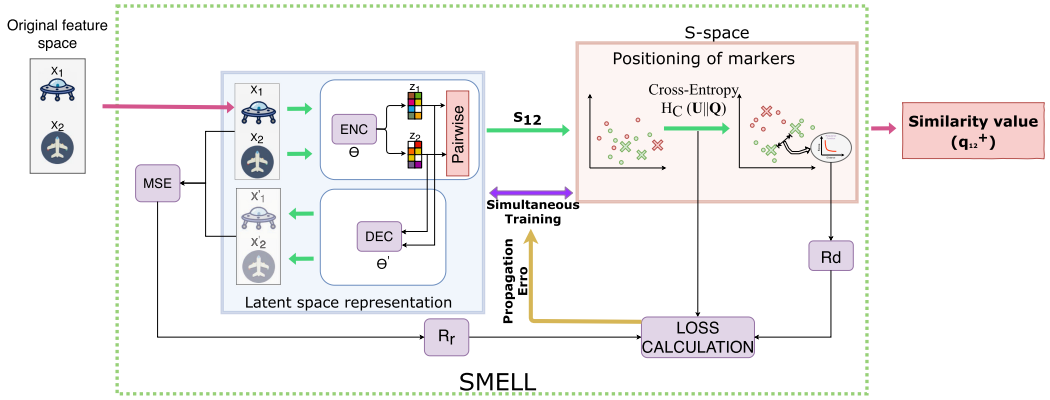


Fig. 3. The left side represents the encoder with reconstruction, and the right side represents the optimization process for the markers' position for $\mathcal{M} = \{\boldsymbol{\mu}_1^+, \boldsymbol{\mu}_2^+, \boldsymbol{\mu}_3^-\}$. In this example, we used two positive markers and one negative marker. Green and red crosses represent $\boldsymbol{\mu}_1^+$, $\boldsymbol{\mu}_2^+$, and $\boldsymbol{\mu}_3^-$, respectively, and green and red dots represent the similar and dissimilar input pairs. The rightmost green arrow represents the markers' position optimization step by using cross-entropy divergence and some regularization functions. Observe that the number of positive and negative markers is a hyperparameter.

So, to minimize the loss function H_c, SMELL tends to approximate the objects in the *S-space* around their respective markers as well as reposition the markers according to S-vectors position—that is, S-vectors are grouped close to the similarity markers; we also observe an analogous behavior to dissimilar S-vectors.

The impact of the attractive behavior is controlled by the constant r_{HC}—that is, the higher the r_{HC}, the greater is the tendency to group the points $\boldsymbol{s}_{ij}$ closer to the respective markers. In addition, note that the regularization functions operate in different spaces—that is, R_r operates in latent feature space, R_d operates in S-space, and H_c operates in latent feature space and S-space simultaneously.

Figure 3 depicts the more detailed schematic of our proposal using a toy example (two positive markers and one negative marker). Observe that the number of positive and negative markers is a hyperparameter.

4.4 Optimization

To find the Σ^* set, we use mini-batch stochastic gradient descent and backpropagation. First, we note that the decoder weights Θ' are only affected by the R_r component of the loss function J. So, we can use $\partial R_r/\partial\Theta'$ to update Θ'. Then, given a mini-batch $(\boldsymbol{x}_i, \boldsymbol{x}_j) \in \mathcal{B}$ with g samples and learning rate λ, Θ' is updated by

$$\Theta' = \Theta' - \frac{\lambda}{g} \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j) \in \mathcal{B}} \frac{\partial R_r(\boldsymbol{x}_i, \boldsymbol{x}_j)}{\partial \Theta'}.$$

To optimize the markers, consider that

$$\boldsymbol{\mu}_t = \boldsymbol{\mu}_t - \frac{\lambda}{g} \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j) \in \mathcal{B}} \frac{\partial J(\boldsymbol{x}_i, \boldsymbol{x}_j)}{\partial \boldsymbol{\mu}_t} = \boldsymbol{\mu}_t - \frac{\lambda}{g} \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j) \in \mathcal{B}} \left(\frac{\partial H_C(\boldsymbol{x}_i, \boldsymbol{x}_j)}{\partial \boldsymbol{\mu}_t} \right) + \frac{\partial R_d}{\partial \boldsymbol{\mu}_t},$$

where $\partial H_C(\boldsymbol{x}_i, \boldsymbol{x}_j)/\partial \boldsymbol{\mu}_t$ can be calculated (as can be seen in the work of Li et al. [25]) for a given $\boldsymbol{\mu}_t$ and $\boldsymbol{s}_{ij}$ as

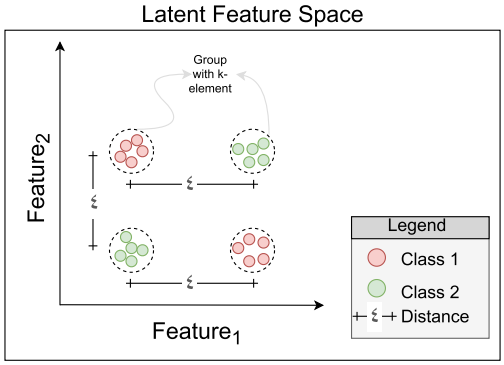


Fig. 4. Toy example for optimal latent space. Same color indicates the same label. S-space requires that each group has only elements of the same class, but note that this space can have different groupings with elements of the same class.

$$\frac{\partial H_C(x_i, x_j)}{\partial \boldsymbol{\mu}_t} = 2\frac{(\boldsymbol{q}_{ij}^t - \boldsymbol{u}_{ij})(\boldsymbol{s}_{ij} - \boldsymbol{\mu}_t)}{1 + ||\boldsymbol{s}_{ij} - \boldsymbol{\mu}_t||_2^2},$$

and

$$\frac{\partial R_d}{\partial \boldsymbol{\mu}_t} = -2 \sum_{\boldsymbol{\mu}_s \in M} \left[\text{sign}(\boldsymbol{\mu}_s) \frac{||\boldsymbol{\mu}_t - \boldsymbol{\mu}_s||_2}{(||\boldsymbol{\mu}_t - \boldsymbol{\mu}_s||_2^2 + \epsilon)^2} \right],$$

where $\text{sign}(\boldsymbol{\mu}_s) = 1$ if $\boldsymbol{\mu}_s \neq \boldsymbol{\mu}_t$ and $\boldsymbol{\mu}_s$ has same semantic (similarity or dissimilarity) than $\boldsymbol{\mu}_t$, and $\text{sign}(\boldsymbol{\mu}_s) = 0$ otherwise. For training SMELL, we randomly selected the mini-batch with m pairs of elements (half are similar, and the other half are dissimilar). In addition, our proposal does not have any specific batch selection criteria, unlike some specialized techniques in the literature [44, 48].

4.5 Theoretical Proprieties

Due to the S-space's construction, we can obtain some theoretical proprieties.

Definition 4.1 (Optimal Latent Space). Let $\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathcal{X}$ and a latent representation function $f_\Theta : \mathcal{X} \rightarrow \mathcal{Z}$. The transformation f_Θ generates an *optimal latent space* $\mathcal{Z}$ when the expected value $\mathbb{E}[||\boldsymbol{s}_{ij}||_2] = 0 \implies l(\boldsymbol{x}_i) = l(\boldsymbol{x}_j)$.

SMELL can group points of the same class into clusters. It is worth noting that we defined the optimal space as a conditional instead of a biconditional statement. From this definition, we can observe that SMELL may create several clusters of the same class, as depicted in Figure 4.

Proposition 4.1. *In S-space, given k positive markers in the set $\mathcal{M}^+$ and $n-k$ negative markers in $\mathcal{M}^-$, if the latent space found by SMELL (i.e., the estimation of the parameters Θ of f_Θ) generates an optimal latent space, then $\exists\ \boldsymbol{\mu}_i \in \mathcal{M}^+$ so that $||\mu_i||_2^2 < ||\mu_j||_2^2$ for any $\boldsymbol{\mu}_j \in \mathcal{M}^-$.*

Proof. The proof for this proposition can be found in Appendix A. □

From Proposition 4.1, if SMELL finds an optimal latent space, at least one positive marker has a smaller norm than the negative marker. In addition, in practice, as we can see in Section 6, at least one positive marker is smaller than all negative markers (the positive marker is located near the origin). We observed that the model builds a latent space of groups with elements of the same class, similar to Figure 4.

Proposition 4.2. *For S-spaces built with one marker in each group, $\mu^+ \in M^+$ and $\mu^- \in M^-$, D^- and D^+ being the Euclidean distance of an object to the negative and positive marker, respectively, the misclassification risk function of a positive marker is derived analytically.*

Proof. The proof can be found in Appendix A. □

Due to the S-space formulation, we obtain the probability of a pair being similar analytically, given the distance of that pair to the positive marker (typically, this probability is estimated, as we can see in the work of Ustinova and Lempitsky [52]).

5 EXPERIMENTAL SETUP

We conducted extensive experiments in several scenarios with different setups to better understand SMELL behavior and effectiveness. Section 5.1 describes the datasets we have employed. Section 5.2 details the classification protocol designed to evaluate our method and the baselines. Section 5.3 discusses the initialization and the architecture of the proposed approach.

5.1 Dataset

5.1.1 General Purpose Datasets. KEEL [51] is an open source[1] Java software tool that can be used for a large number of different knowledge data discovery tasks. We used 28 datasets provided by KEEL to evaluate our proposal. All datasets are numeric and have no elements missing. Furthermore, all datasets have been min-max normalized to the interval [0, 1], a precondition to the experiments' execution. There is a wide variety of data in KEEL. The 28 datasets used in our experiments are divided into medical data (Bupa, Cleveland, Appendicitis, Newthyroid, Pima, Wdbc, Wisconsin, and Thyroid), natural Language processing data (Vowel, Letter, and Phoneme), experimental psychological data (Balance), feature-based image data (Magic, and Satimage), hierarchical decision-making data (Monk-2, Ring, and Twonorm), nature data (Iris and Banana), disaster prediction data (Titanic), weather data (Ionosphere), chemical data (Glass, Wine, Winequality-red), and object/shape recognition (Sonar, Movement_libras, and Vehicle).

These 28 datasets have substantial diversity in terms of data factors: the number of examples, the number of features, and the number of classes. Specifically, the number of examples ranges from 106 to 2,003, and the number of features ranges from 2 to 90. The datasets contain binary and multiple class datasets with a maximum of 26 classes for one dataset. Although our method scales up to large datasets, some methods do not; hence, due to a large number of datasets, we downsampled some of them (the ones with more than 1,000 samples) to 10% of the original size.

5.1.2 Image Dataset. The MNIST dataset[2] is one of the most common datasets used for image classification and is accessible from many different sources. The dataset consists of grayscale images with 28 x 28 dimensions. Following the work of Hadsell et al. [17], the training set is built from all handwritten digits 4 and 9 from the MNIST dataset. Due to a large number of MNIST features, the spatial correlation found in the images, and a large number of samples, we consider the dataset suitable for this evaluation. All images were normalized to the interval [0, 1], resulting in 6,958 and 6,824 images corresponding to handwritten digits 4 and 9, respectively. In addition, we evaluated our approach with Fashion MNIST and CIFAR10 datasets.

5.2 Network Evaluation

To evaluate all MeL techniques assessed in this work, including our approach, we apply a KNN classifier, with three neighbors, in agreement with Deng et al. [11]. The KNN classification

[1] https://sci2s.ugr.es/keel/studies.php?cat=clas.
[2] http://yann.lecun.com/exdb/mnist/.

performance can often be significantly improved through (supervised) MeL. In this work, the KNN classification can be exchanged for any other algorithm that uses a metric.

Since we used several datasets to validate our proposal, we divided our assessment into two approaches. The first is an individual evaluation for each dataset, and the second is a general evaluation for all datasets.

For each dataset, we calculate the accuracy. For all datasets (except the imsgens datasets), we calculate the average accuracy (Accuracy_AVG), the average rank position value (Ranking_AVG), and the difference of the accuracy average for the best proposal (Diff_AVG). We also calculated the number of times our algorithm was in the first position (# of 1st). For the MNIST dataset, we evaluated the proposals' accuracy for different latent space representation dimensions. We used 10-fold cross validation. This validation can largely retain heterogeneous distributions in the training set and improve statistical confidence in the results. For the sake of reproducibility, our proposal is publicly available on a GitLab repository.[3]

5.3 Parameters Initialization and Network Architecture

We initialize all weights of the autoencoder layers from a zero-mean normal distribution $\mathcal{N}(\mu = 0, \sigma = 0.01)$. Biases were also initialized as outcomes of a normal distribution $\mathcal{N}(\mu = 0.5, \sigma = 0.01)$, following Koch et al. [24]. Markers' positions are initialized with Lloyd's algorithm [29]. Furthermore, we pre-trained an autoencoder (without markers) and further transferred the learn to the complete model (with markers) to improve the convergence speed.

The encoder of all DMeL approaches used as a baseline is identical to the one we used in SMELL. According to Xie et al. [65], we set network dimensions to m-512-512-2048-n for all datasets, where m is the number of features of the input data, and n is the latent space representation dimension. All layers are fully connected, and we used as an activation function the rectified linear unit (ReLU) [35]. In addition, we used mini-batch stochastic gradient descent, where the learning rate is 0.01. All parameters mentioned previously (except for the calibration of the markers) were used in all DMeL baselines and our proposal. Since the optimization model depends on some hyperparameters (r_{HC}, r_d, r_r, w, k), we performed an investigation to determine which value of these variables would maximize the model accuracy. Therefore, we randomly chose the Vehicle dataset to train the model and select the hyperparameters.

We use Bayesian optimization for hyperparameters tuning. Because some hyperparameters are defined in a discrete interval, such as the number of markers, it was necessary to perform the discretization of the Bayesian optimization values. We used five random starting points and then 20 rounds of the algorithm, where we found $r_{HC} = 1$, $r_d = 10^{-1}$, $r_r = 10^{-3}$, similarity markers $k = 3$, and dissimilarity markers $w - k = 2$; these values were used in the rest of this work. We realized that our proposal typically performs well when k has a value similar to $w - k$.

We configured all baselines with the hyperparameters recommended in their original articles. These parameters are listed next; observe that three approaches (Euclidean, NCA, and NPair) do not have hyperparameters to set:

- MeL algorithm
  - *ANMM* [55]: The size of the homogeneous and heterogeneous neighborhoods for each data point is set to 10.
  - *KDMLMJ* [38]: Let k_1, k_2 denote the number of neighbors for constructing the positive and negative difference spaces; we used $k_1 = k_2 = 5$.

[3] https://gitlab.com/sufex00/smell.

Table 2. Performance Comparison Summary of an Ablation Study When Using KNN Classification for 27 Different Datasets

Proposal	Accuracy_AVG	Ranking_AVG	Diff_AVG	# of 1st
SMELL ($r_r = 0$)	0.8254	**2.2857**	0.0116	9
SMELL ($r_d = 0$)	0.8169	2.8571	0.0201	5
SMELL ($r_r = r_d = 0$)	0.8178	**2.2857**	0.0193	9
SMELL (Euclidean)	0.7994	3.6429	0.0376	4
SMELL (S-space)	**0.8268**	**2.2857**	**0.0102**	**10**

The full table can be seen in Appendix B.

- DMeL algorithm
  - *ContrastiveLoss* [17]: The margin term equals 1.
  - *Triplet* [44]: The margin term equals 0.2.
  - *MultiSimilarityLoss* [59]: The hyperparameters for the model are $\alpha = 2$, $\lambda = 1$, and $\beta = 50$.
  - *FastAPLoss* [3]: The number of soft histogram bins for calculating average precision is 10.

6 RESULTS AND DISCUSSION

In this section, we present the results of the SMELL's assessment. We also discuss the interpretability of the S-space and conduct a performance evaluation comparing SMELL with three distance MeL approaches from pyDML [49][4]: ANMM [55], NCA [15], and KDMLMJ [38]; five DMeL approaches: Contrastive [17], Triplet [44], NPair [48], FastAP [3], and MSLoss [59]; and Euclidean distance.

6.1 Ablation Study

For a better understanding of our proposal, we conducted an ablation study. Therefore, we evaluated SMELL for different regularization calibration values. We evaluated SMELL with and without the reconstruction error ($r_r = 0$), with and without the repulsive error ($r_d = 0$), and without both ($r_d = r_r = 0$). The other default values adopted in our experiments can be found in Section 5.

In addition, we evaluated the behavior of our proposal when using S-space only for training. We then use a version of SMELL without the S-space (using Euclidean distance). We named this approach *SMELL (Euclidean)*. Therefore, after training the model using S-space, we observe only the latent space to perform the similarity metric's extraction—that is, we consider that the similarity between two objects is the Euclidean distance between them in the latent space. It is also worth noting that we use the same default values adopted in our experiments (without any restriction on r_d and r_r). A summary of results is shown in Table 2. We also provide a complete report of our results in Table B.1 (Appendix B).

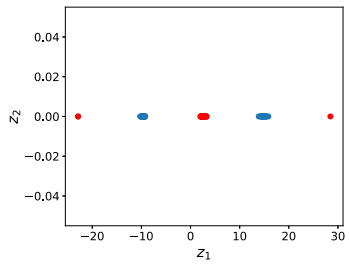
We see that among the usual SMELL methods when we take $r_d = 0$, the proposal tends to have performance degradation. This behavior is easily seen in the dataset ring (see Table B.1 in Appendix B), in which SMELL ($r_d = 0$) and SMELL ($r_d = r_r = 0$) have an accuracy of 0.6536 and 0.6610, respectively. Comparing this value with the best result, we have a difference of more than 20%. The proposal with $r_d = 0$ has the worst performance in all four metrics analyzed (excluding Euclidean).

When $r_r = 0$, we observe a slight impact on the result (when compared to $r_d = 0$), but for the datasets Appendicitis, Vowel, Banana (10%), and Twonorm (10%) (see Table B.1 in Appendix B), changing r_r to zero made SMELL stop being the first position (when analyzing accuracy), moving to the second to last position.

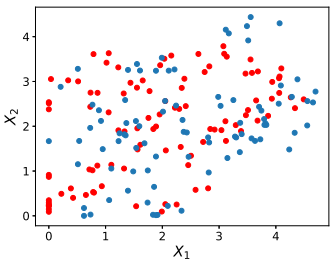
[4]https://pydml.readthedocs.io/en/latest/index.html.

Table 3. Performance Comparison Using Different S-space Dimensions

Dimension	ACC		
	Fashion-MNIST	MNIST	CIFAR10
$n = 16$	0.8392	0.9919	0.7504
$n = 32$	0.8494	0.9912	0.7698
$n = 64$	**0.8757**	**0.9925**	**0.7892**
$n = 128$	0.8564	0.9922	0.7713
$n = 256$	0.8416	0.9922	0.7787



(a) Latent space ($n = 2$) from SMELL



(b) Encoder output ($n = 2$) without markers

Fig. 5. Sonar dataset: latent space (a) and encoder space (b) analysis.

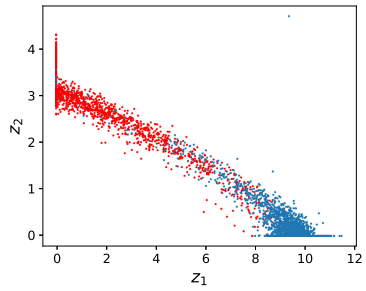
When we consider the case of SMELL with the Euclidean metric (instead of S-space), our proposal has the worst performance among the cases analyzed for the four metrics adopted. In particular, we see that the average difference for the first place (DIFF_AVG) has increased 200%. In addition, it is worth noting that for the datasets Twonorm (10%), Banana (10%), Wdbc, Movement_libras, and Appendiciteps (see Table B.1 in Appendix B), SMELL goes from first place to last place. This evidences the limitation of the Euclidean metric (even using the function f_Θ found by our proposal).

We hypothesized that SMELL tends to find a representation in S-space that captures similarity semantics. SMELL (S-space) has the best performance among all the versions used, which is evidence of the last statement. In addition, a repulsive regularizer tends to increase the separability of latent space, which shows the importance of the repulsive regularizer.

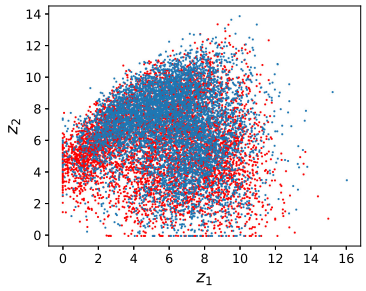
To perform a sensitivity analysis in the S-space, we conducted an experiment varying the S-space dimension using the same experiment design presented in the article. We observe that $n = 64$ presents the best overall performance, and the results are shown in Table 3.

Moreover, we evaluated the behavior of SMELL in comparison with an autoencoder (without markers). Figure 5 shows the behavior of the encoder output under SMELL and the autoencoder. In Figure 5(a), whose encoder was used with SMELL, the classes have well-defined groups, differently from Figure 5(b), where there is a greater dispersion of the classes, with no clustering pattern being observed. The same behavior is found in the MNIST dataset, as shown in Figure 6.

Finally, we investigated the behavior of the similarity function found by SMELL. Precisely, we randomly select an element $\boldsymbol{x}_i$ and calculate the similarity using the SMELL against all similar pairs $(\boldsymbol{x}_i, \boldsymbol{x}_j)$—that is, $\boldsymbol{x}_i$ has the same label as $\boldsymbol{x}_j$. We selected a single object x_i for each evaluation among the classes. We can see in Figure 7(a) the distribution of the similarity (q^+) for the single random class. We can see that the distribution is asymmetrical, with a tail to the left and the median is 0.7887. This behavior indicates that most points have high similarities. In addition, we see that the mean is 0.8039, even in a heavy tail distribution, which tends to decrease the mean

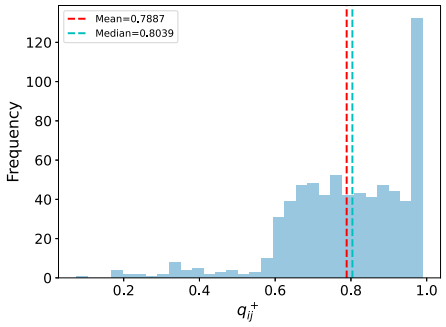


(a) Latent space ($n = 2$) from SMELL

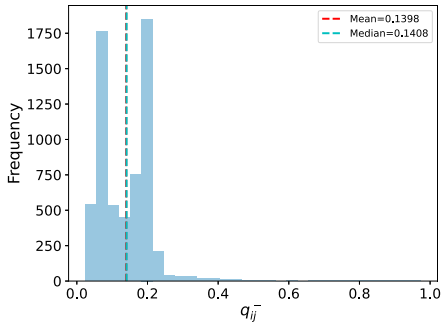


(b) Encoder output ($n = 2$) without markers

Fig. 6. MNIST dataset: latent space (a) and encoder space (b) analysis.



(a) Similarity distribution



(b) Dissimilarity distribution

Fig. 7. FASHION-MNIST train. (a) The class similarity distribution for a random element. (b) The class dissimilarity distribution for a random element.

considerably. Figure 7(b) shows similar behavior for an element of another class. The distribution of the dissimilarity (q^{-}) as median equals 0.1398 and median equals 0.1408, and we also can see a heavy tail to the right. These results corroborate that SMELL consistently groups similar objects once the similarity of same-class objects is high.

6.2 Performance Comparison

To compare our results to other techniques present in the literature, we used the datasets and the metrics appointed in Section 5, with $n = 64$ (latent dimension). The summary of results can be found in Table 4 (for complete results, see Appendix C). The second column indicates whether the approach is based on MeL or DMeL techniques. We compare SMELL to MeL approaches [15, 38, 55], DMeL approaches [3, 17, 44, 48, 59], and the usual Euclidean distance. The k-fold cross-validation results are shown by averaging the standard deviation and accuracy values reported by the process.

SMELL achieved the best accuracy results in 7 (# of 1st) datasets, thus surpassing all other analyzed algorithms (improving 40% more datasets when compared with the second best). KDMLMJ and MSLoss, the second-best, achieved the best result in five datasets. SMELL achieved an accuracy of 0.8268 (Accuracy_AVG). The second-best, Triplet, achieves 0.8115, and the third-best, MSLoss, achieves 0.8081. In a simple dataset (Monk-2), SMELL achieves 100%. It is worth mentioning that SMELL—even in some situations its performance is not the best—reaches accuracy close to the best algorithm. For instance, the average distance between SMELL and the best algorithm is 2.26% (Diff_AVG), improving its average distance by 67.70% and 84.96% compared to Triplet (second best)

Table 4. Performance Comparison of Some Distance Metrics Approaches and SMELL When Using KNN Classification for 27 Different Datasets

Proposal	Category	Accuracy_AVG	Ranking_AVG	Diff_AVG	# of 1st
ANMM	MeL	0.7768	5.7143	0.0726	0
KDMLMJ	MeL	0.7824	4.9643	0.0667	5
Contrastive	DMeL	0.6923	7.1786	0.1571	1
MSLoss	DMeL	0.8081	3.7857	0.0418	5
Triplet	DMeL	0.8115	3.9643	0.0379	4
NCA	MeL	0.7732	5.6786	0.0762	3
NPair	DMeL	0.7380	6.2143	0.1114	2
FastAP	DMeL	0.7801	4.7857	0.0693	3
Euclidean	-	0.7486	5.8929	0.1007	1
SMELL	DMeL	**0.8268**	**3.6429**	**0.0226**	**7**

The best results are in **bold**.

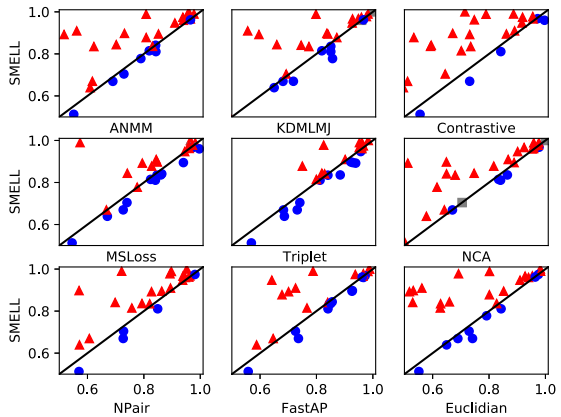


Fig. 8. Comparison between SMELL and all proposals used as a baseline. The red triangles, blue dots, and gray squares indicate when the SMELL is superior, inferior, and has the same mean accuracy result, respectively, when compared to other proposals.

and MSLoss (third best), respectively. Finally, when we average the ranking, SMELL achieved an average of 3.6429 (Ranking_AVG), the smallest value among all algorithms. The second best was MSLoss, reaching 3.7857.

In Figure 8, we compare SMELL's accuracy with all other approaches used in this article. We noticed that SMELL, in all cases, manages to overcome the techniques presented when we compare the number of individual hits—that is, the number of datasets that SMELL exceeds the accuracy of the analyzed baseline. In addition, we noticed a small scattering of the blue dots around the black line compared to the red triangles' behavior. It indicates that its results are close to the best even when SMELL performs worse than another approach.

Analyzing the MeL approaches only (see Table 4), we see that KDMLMJ and NCA algorithms achieve better results (among the algorithms adopted as baseline) when considering the metric that counts the number of times that the algorithm's accuracy surpassed all of the others. This behavior is because the algorithms have been evaluated with KNN, and these algorithms were specifically designed to improve this classifier.

Table 5. Performance Comparison of Some Distance Metrics Approaches and SMELL When Using KNN Classification for Two Different Image Datasets

Proposal	MNIST (only 4 and 9)	Fashion MNIST	CIFAR10
Contrastive	0.7945	0.7573	0.7211
MSLoss	0.9804	0.8420	0.7754
Triplet	0.9906	0.8724	0.7469
NPair	0.9889	0.7624	0.6956
FastAP	0.9911	0.8607	0.7840
SMELL	**0.9925**	**0.8757**	**0.7892**

The best results are in **bold**.

Finally, we evaluated our proposal with three image datasets (Table 5). We verified the consistency of SMELL for the datasets analyzed—that is, for MNIST, we obtained the best result of 0.9925 accuracy. Compared to the other approaches, FastAP loss had the second-best result, whereas Triplet loss had the third-best result. For the Fashion MNIST dataset, we verified that SMELL and Triplet obtained the best and second-best results, respectively. We conducted an investigation into CIFAR10, and we performed the best result.

6.3 Behavior Analysis

Our proposal is based on optimizing the parameter set $\Sigma = \{\Theta, \Theta', \boldsymbol{M}\}$ using markers (with a Cauchy kernel). SMELL learns a representation of input pairs that groups the points with similar and dissimilar labels around their respective markers. We can observe this behavior in Figure 9. In this figure, the input pairs of similar and dissimilar labels are represented by pink circles and gray triangles. In addition, $\boldsymbol{\mu}^+$ and $\boldsymbol{\mu}^-$ markers are represented by green and red crosses, respectively. We plot some $\boldsymbol{s}_{ij}$ vectors for input pairs $(\boldsymbol{x}_i, \boldsymbol{x}_y)$ of the test set for the Balance dataset. After training the autoencoder, a two-dimensional plot was created with the aid of PCA before (first figure) and after (the other figures) the optimization process.

The points do not present a well-defined cluster structure in the first plot in Figure 9 (before adjusting the markers' positions). This behavior changes when we analyze the last plot in Figure 9 (after adjusting the markers). In the last plot in Figure 9, we can see well-defined groups around the markers. Moreover, by comparing the scale of the Figures 9, we see that in the last case, points are more spaced—that is, our proposal tends to group points around their respective markers. This behavior corroborates our initial hypothesis described in hypothesis 1 (H1).

We observed that our proposal acts as an attractive potential. In this sense, the marker "pulls" the favorable points (the similarity mark "pulls" similar points). Therefore, their movement resembles a group mobility model [19]—that is, the marker is being positioned, and the points go "following" the leader as a "caravan" of nomads. At the same time, the markers tend to repulse themselves.

To quantify the clustering of vectors around a reference point (e.g., centroid), we can use the **Mean Squared Distance (MSD)**. So, we calculate the MSD during the training of our proposal for each S-vector to its respective marker—that is, the Euclidean distance between similar/dissimilar s_{ij} and similarity/dissimilarity marker. In addition, we also calculate loss function j, as shown in Figure 10(a). We observed that as loss decreases, the MSD also decreases, indicating that, during training, our model groups s_{ij} close to their respective markers. This can be seen as such an intense attracting field, which locks the movement dynamics of the points closest to the markers.

Figure 10(b) depicts the training duration of the DMeL proposals in the Fashion-MNIST dataset (63,000 train samples). DMeL has similar training times because the encoder function (NN) training is the primary time bottleneck. Regarding the prediction's time, similarity metric extraction

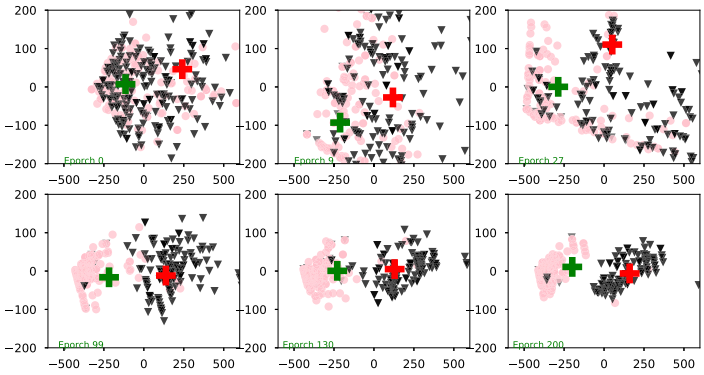
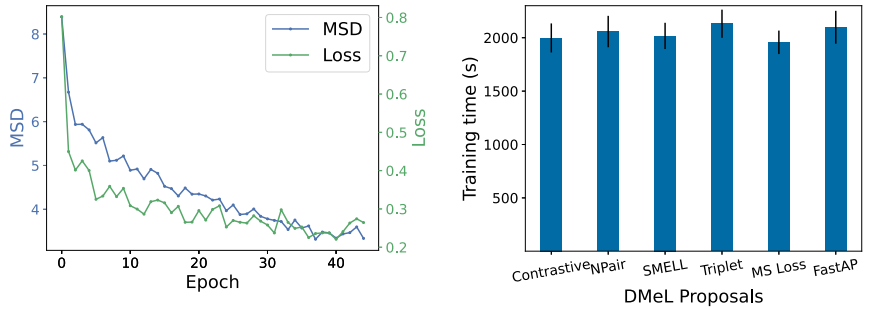


Fig. 9. Simultaneous training of μ^+ (green) and μ^- (red) markers' position and data representation in S-space for some training epochs.



(a) Loss and Mean Squared Distance (b) Training time

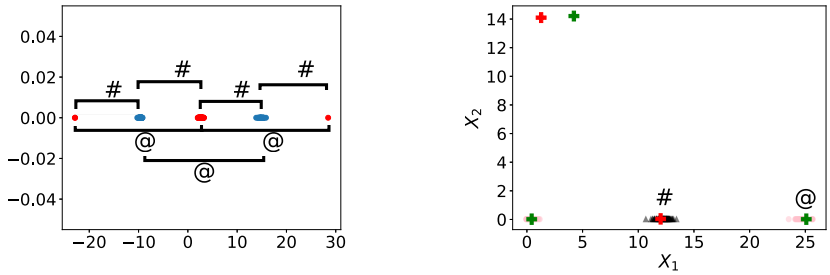
Fig. 10. FASHION-MNIST train. (a) Loss and MSD (between S-vector and marker) for each train epoch. (b) DMeL proposals training time for 10 rounds.

proposals use based-metrics algorithms (e.g., KNN) to perform inference, as we can see in other works [3, 44, 48]. Thus, some heuristics can improve the proposal's prediction time effectiveness, as seen in the work of Schroff et al. [44]. For example, the time prediction for the Fashion-MNIST dataset (7,000 test samples) is 17.39 seconds.

6.4 Latent Space and S-space Analysis

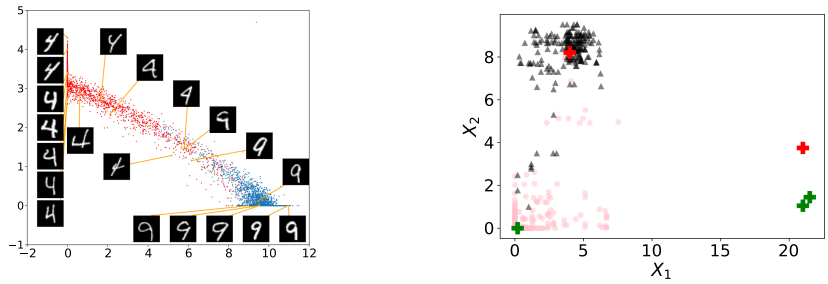
For a better understanding of the latent space found by SMELL, we analyzed the behavior of our proposal using the sonar and MNIST datasets as shown in Figures 11 and 12. For the sake of visualization, in this analysis, we use the setup discussed in Section 5 with $n = 2$ (latent dimension). Figures 11(a) and 12(a) show the latent feature space (output of encoder). Observe that in these figures, points represent individual objects. Red and blue points represent different classes. There are two classes in the sonar dataset, and we show only two classes of MNIST (handwritten digits 4 and 9). These latent feature spaces result from the joint optimization process of the autoencoder and the S-space.

In Figure 11(a), we observe that red points are grouped in different regions far apart at a distance approximately constant, denoted by @. Similarly, blue points are apart at a distance approximately constant, @. Different clusters are apart at a distance approximately constant, denoted as #.



(a) Latent space ($n = 2$) (b) S-space ($n = 2$)

Fig. 11. Sonar datasets: latent space (a) and s-space (b) analysis for SMELL.



(a) Latent space ($n = 2$) (b) S-space ($n = 2$)

Fig. 12. MNIST datasets: latent space (a) and S-space (b) analysis for SMELL.

Figures 11(b) and 12(b) show a random sample of 400 data pairs from the sonar dataset mapped to S-space (200 similar and 200 dissimilar pairs). In S-space, points represent a pair of objects. Pink circles and black triangles represent similar and dissimilar labels, respectively. In addition, similarity and dissimilarity markers are represented by green and red crosses, respectively.

Figure 11(b) shows some clustered regions. The region grouped by the similarity marker (closer to the origin) is responsible for grouping elements of similar classes with a distance closer to 0. This result corroborates with Proposition 4.1. The same behavior is found in Figure 12(b), where we observe a green cross close to the origin.

However, in Figure 11(b), we observe some similar objects mapped to points with a distance close to @, instead of zero. The green cross at @ is responsible for creating the similarity region that represents this situation. Depending on the data complexity, other regions of similarity and/or dissimilarity can occur and are represented by other green/red crosses. Dissimilarity regions are depicted as #. Therefore, in the space found by SMELL, we see the behavior of multiple groups, separated by distances determined by the similarity/dissimilarity markers (labels # and @).

Observe in Figure 12(a) the soft transition from digits 4 to 9, which shows that the S-space preserves the connection between these two similar digits. This effect is captured even though we do not use any data-specific feature extractor, such as convolution layers. In SMELL, the encoder can be switched by any feature extractor (e.g., vision transformer) tailored explicitly for the input data.

In Figure 12(a), we observe that the handwritten digits four are grouped (on the left). Even in this group, the similarities between the digits remain. The first two digits in the top-left region correspond to numbers with thicker writing and slightly rotated, and as we go down in the latent

space, the digit's shape starts to become thinner. This behavior indicates a gradient that represents the thickness of the object. This same behavior occurs similarly to digit 9. There is a transition from groups of digits 4 to 9—that is, there is a semantic in this transition. As we gradually move along the diagonal that connects the two groups, digits 4 resemble 9, so it is tough to differentiate between them in the middle of the diagonal. It is also worth noting that the farther away from the denser regions of the points cloud, the less readable the numbers are. For instance, the two digits four are depicted below the transition diagonal. We see that our proposal uses markers to help in the convergence and finds a latent space that preserves the semantics of the original data. This behavior corroborates our initial hypothesis described in hypothesis 2 (H2).

It is also worth noting that our proposal has no sensitive learning in the presence of multiple markers—that is, even in this experiment where we have defined three similarity markers and two dissimilarity markers, our proposal does not use all. This behavior is emphasized in Figures 11(b) and 12(b), where our proposal removes excessive markers from the groupings by locating these markers far away from the data, and this behavior indicates that the number of markers is a virtual parameter of the model.

We hypothesize that markers group data points considered similar (in our context, which have the same labels) and dissimilar (different labels) in disjoint regions. Figures 12(b) and 11(b) show this behavior, where we can see similar and dissimilar groups in distinct (and disjoint) regions in S-space. In addition, we can notice in Figure 11(a) that our proposal allows different clusters for the same class (optimal latent space), as mentioned in Definition 4.1.

7 CONCLUSION

In this work, we proposed SMELL, based on the fact that the distance metrics can be simultaneously learned along with a latent representation of the data and the similarity markers. We hypothesized that SMELL groups data points consider similar and increases class separability. We showed evidence that supports our hypothesis through comprehensive behavior analysis. We also extensively validated our proposal by comparing it to many methods over different types of input data. We obtained promising results, and, in general, we got the best results. We intend to investigate the possible applications for this type of approach and use Proposition 4.2 to build a novel loss function specifically tailored for SMELL.

APPENDICES

A PROOFS

Proposition A.1 (4.1). *In S-space, given k positive markers in the set $\mathcal{M}^+$ and $n-k$ negative markers in $\mathcal{M}^-$, if the latent space found by SMELL (i.e., the estimation of the parameters Θ of f_Θ) generates an optimal latent space, then $\exists\ \boldsymbol{\mu}_i \in \mathcal{M}^+$ so that $||\mu_i||_2^2 < ||\mu_j||_2^2$ for any $\boldsymbol{\mu}_j \in \mathcal{M}^-$.*

Proof. Given $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, SMELL measures the similarity between the entries through the Cauchy kernel given by q_{ij}^+ so that for Σ^* it follows that $q_{ij}^+ = 1 \iff l(x_i) = l(x_j)$. Since f_Θ generates an optimal space, we then have $\mathbb{E}[||s_{ij}||_2] = 0 \implies l(x_i) = l(x_j)$, so it follows that for a optimal latent space, we must have

$$q_{ij}^+ = \frac{\sum_{k\in\mathcal{M}^+}(1+||\mu_k||_2^2)^{-1}}{\sum_{k\in\mathcal{M}^+}(1+||\mu_k||_2^2)^{-1} + \sum_{s\in\mathcal{M}^-}(1+||\mu_s||_2^2)^{-1}} = \frac{1}{1+\frac{\sum_{s\in\mathcal{M}^-}(1+||\mu_s||_2^2)^{-1}}{\sum_{k\in\mathcal{M}^+}(1+||\mu_k||_2^2)^{-1}}}.$$

Hence, if we want $l(x_i) = l(x_j)$, we should ideally have q_{ij}^+ tends to 1. It follows that $\sum_{s\in\mathcal{M}^-}(1+||\mu_s||_2^2)^{-1} < \sum_{k\in\mathcal{M}^+}(1+||\mu_k||_2^2)^{-1}$. Therefore, let μ^+ be the element with the smallest module in the

set $\mathcal{M}^+$; we then have $\sum_{k\in\mathcal{M}^+}(1+||\mu_k||_2^2)^{-1} < k(1+||\mu^+||_2^2)^{-1}$. Analogously, we can consider μ^- as the vector with the largest module in the set $\mathcal{M}^-$, so $\sum_{k\in\mathcal{M}^-}(1+||\mu_k||_2^2)^{-1} > (n-k)(1+||\mu^-||_2^2)^{-1}$.

We can then conclude that $k(1+||\mu^+||_2^2)^{-1} > (n-k)(1+||\mu^-||_2^2)^{-1}$, and therefore $(n-k)(1+||\mu^+||_2^2) < k(1+||\mu^-||_2^2)$. Furthermore, adding the restriction that SMELL has a similar count of positive and negative markers (Section 5.3), we have $1+||\mu^+||_2^2 < 1+||\mu^-||_2^2$. □

Proposition A.2 (4.2). *For S-spaces built with one marker in each group, $\mu^+ \in M^+$ and $\mu^- \in M^-$, the misclassification risk function of a positive marker is derived analytically.*

Proof. First, we consider the risk of the similarity in a random negative pair to be more than the similarity in a random positive pair [52] as $R = \int_{-\infty}^{\infty} p^-(x)[\int_{-\infty}^{y} p^+(y)dy]dx$. Consider $\mathcal{M}^+ = \{\mu^+\}$ and $\mathcal{M}^- = \{\mu^-\}$—that is, $\mathcal{M}^+$ and $\mathcal{M}^-$ have cardinality 1, and $D^+(x_i,x_j)$ and $D^-(x_i,x_j)$ are the Euclidean distances from the point s_{ij} to the positive and negative markers, respectively. The risk of misclassification is

$$R^+ = \int_0^{D^+(x_i,x_j)} \left[p^-(D^+(x_i,x_j)) \left(\int_0^{y=D^+(x_i,x_j)} p^+(y)d(y) \right) d(D^+(x_i,x_j)) \right].$$

Therefore, due to the construction of S-space, we consider that the likelihood of similarity/dissimilarity between the s_{ij} representation of two samples is calculated as the relative distance to a marker. Due to this construction, we have $R^+ = \int_0^{D^+(x_i,x_j)} q^-[\int_0^{y=D^+(x_i,x_j)} q^+d(y)]d(D^+(x_i,x_j))$. Calculating each term separately, we have that for the markers in the sets $\mathcal{M}^+$ and $\mathcal{M}^-$, the probability of a having a similar objects is

$$q^+ = \frac{(1+D^+(x_i,x_j)^2)^{-1}}{(1+D^+(x_i,x_j)^2)^{-1}+(1+D^-(x_i,x_j)^2)^{-1}} = \frac{1+D^-(x_i,x_j)^2}{2+D^+(x_i,x_j)^2+D^-(x_i,x_j)^2}.$$

Analogously, we can find that $q^- = 1-q^+$. To simplify the notation, consider that D^+ equals $D^+(x_i,x_j)$ and D^- equals $D^-(x_i,x_j)$. We have for $R^+ = \int_0^{D^+}(1-q^+)[\int_0^{y=D^+} q^+d(y)]d(D^+)$. Therefore, the risk of misclassification for the positive marker can be reduced to $R^+ = \int_0^{D^+}(1-q^+)\Phi(D^+)dD^+$, where Φ is the cumulative density function for q^+. Calculating the accumulated histogram $\Phi(x) = \int_0^x \frac{1+(D^-)^2}{2+(x)^2+(D^-)^2}dx$, we get $\Phi(x) = \frac{((D^-)^2+1)tan^{-1}(\frac{x}{\sqrt{(D^-)^2+2}})}{\sqrt{(D^-)^2+2}}$. Therefore, by solving the integral $\int_0^{D^+}(1-q^+)\Phi(D^+)dD^+$, we have the exact analytical value of the misclassification risk function for the positive marker. Finally, it follows that

$$R^+ = \frac{(D^-)^2+1}{\sqrt{(D^-)^2+2}}\left[\left(\sqrt{(D^-)^2+2}\right) log\left(\frac{1}{\sqrt{\frac{(D^+)^2}{(D^-)^2+2}+1}} \right) - \frac{((D^-)^2+1)\left[tan^{-1}\left(\frac{(D^+)}{\sqrt{(D^-)^2+2}}\right)\right]^2}{\sqrt{(D^-)^2+2}} + (D^+)tan^{-1}\left(\frac{(D^+)^2}{\sqrt{(D^-)^2+2}}\right)\right].$$ □

B ABLATION RESULTS

This appendix show a full comparison of an ablation study for 27 different datasets (Table B.1).

Table B.1. Performance Comparison of an Ablation Study When Using KNN Classification for 27 Different Datasets

Dataset	SMELL ($r_r = 0$)	SMELL ($r_d = 0$)	SMELL ($r_r = r_d = 0$)	SMELL (Euclidean)	SMELL (S-space)
Appendicitis	78.90 ± 11.12	79.09 ± 11.02	79.09 ± 09.59	77.18 ± 9.13	**80.19 ± 7.74**
Balance	97.00 ± 01.03	97.00 ± 01.12	97.00 ± 1.02	98.40 ± 1.22	**98.88 ± 1.02**
Banana (10%)	89.44 ± 4.89	90.76 ± 08.83	90.01 ± 3.76	87.73 ± 1.14	**90.95 ± 4.42**
Bupa	64.58 ± 09.92	**67.05 ± 09.62**	66.37 ± 8.83	55.15 ± 10.12	63.92 ± 6.85
Cleveland	**52.22 ± 07.42**	51.20 ± 05.65	52.21 ± 6.32	51.93 ± 7.22	51.24 ± 8.49
Glass	67.52 ± 14.71	67.81 ± 12.16	66.91 ± 11.31	**70.75 ± 11.46**	66.94 ± 13.24
Ionosphere	**89.75 ± 5.14**	88.89 ± 02.37	**89.75 ± 4.98**	84.88 ± 6.79	89.47 ± 4.59
Iris	95.33 ± 04.00	94.67 ± 04.47	94.67 ± 4.27	95.33 ± 4.27	**96.00 ± 3.26**
Letter (10%)	**80.30 ± 3.72**	77.58 ± 08.40	77.77 ± 3.82	62.96 ± 8.62	77.77 ± 4.72
Magic (10%)	84.44 ± 02.98	84.44 ± 02.74	**84.49 ± 2.37**	78.92 ± 7.11	83.49 ± 2.58
Monk-2	**100 ± 0.00**	**100.00 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**
Movement-libras	86.21 ± 04.44	84.72 ± 04.44	85.56 ± 5.39	83.33 ± 6.92	**87.78 ± 3.61**
Newthyroid	**97.71 ± 02.25**	**97.71 ± 03.57**	96.77 ± 2.29	90.71 ± 10.97	96.77 ± 2.11
Phoneme (10%)	82.77 ± 03.89	**83.87 ± 04.97**	82.21 ± 3.04	81.29 ± 4.89	81.48 ± 4.31
Pima	70.44 ± 05.17	70.98 ± 05.96	**71.75 ± 2.65**	69.68 ± 3.87	70.44 ± 3.52
Ring (10%)	**89.32 ± 21.56**	65.36 ± 19.21	66.10 ± 3.11	71.80 ± 16.55	89.22 ± 4.56
Satimage (10%)	83.49 ± 02.94	85.20 ± 03.12	84.58 ± 4.45	**85.37 ± 3.11**	84.11 ± 3.57
Segment (10%)	**90.48 ± 04.48**	89.05 ± 07.22	89.05 ± 4.42	90.27 ± 4.76	89.76 ± 3.96
Sonar	84.05 ± 10.53	**85.52 ± 10.92**	84.55 ± 10.44	81.76 ± 12.88	83.59 ± 11.26
Thyroid (10%)	95.59 ± 02.43	95.56 ± 02.15	**95.69 ± 2.51**	94.71 ± 2.24	94.71 ± 2.45
Titanic (10%)	62.41 ± 16.07	62.56 ± 05.23	63.76 ± 16.22	**73.13 ± 6.13**	66.97 ± 15.85
Twonorm (10%)	96.00 ± 01.41	96.75 ± 06.07	97.00 ± 1.48	93.78 ± 13.33	**97.43 ± 1.76**
Vehicle	84.40 ± 02.00	83.34 ± 12.02	**84.87 ± 3.33**	78.32 ± 18.33	84.40 ± 2.63
Vowel	98.12 ± 00.79	98.12 ± 15.63	**98.99 ± 1.09**	98.48 ± 1.04	**98.99 ± 0.90**
Wdbc	**96.65 ± 02.99**	96.47 ± 02.40	**96.65 ± 2.99**	89.11 ± 14.63	**96.65 ± 2.89**
Wine	97.77 ± 03.59	97.71 ± 05.12	**98.30 ± 5.11**	97.19 ± 5.14	97.77 ± 5.11
Wisconsin	**96.21 ± 02.14**	95.91 ± 01.84	95.62 ± 1.87	95.91 ± 2.48	**96.21 ± 2.26**
Accuracy_AVG	0.8254	0.8169	0.8178	0.7994	**0.8268**
Ranking_AVG	**2.2857**	2.8571	**2.2857**	3.6429	**2.2857**
Diff_AVG	0.0116	0.0201	0.0193	0.0376	**0.0102**
# of 1st	9	5	9	4	**10**

C PERFORMANCE COMPARISON

This appendix show a full comparison of all DMeL using in this work (Table C.1).

Table C.1. Performance Comparison of Some Distance Metrics Approaches and SMELL When Using KNN Classification for 27 Different Datasets

Dataset	ANMM	KDMLMJ	Contrastive	MSLoss	Triplet	NCA	NPair	FastAP	Euclidean	SMELL
Appendicitis	84.27 ± 10.64	**85.09 ± 9.94**	84.18 ± 9.95	84.09 ± 11.02	81.27 ± 10.79	84.09 ± 9.39	84.90 ± 08.61	84.09 ± 10.07	84.27 ± 10.64	81.09 ± 7.74
Balance	80.81 ± 4.22	79.83 ± 04.32	78.93 ± 23.54	98.24 ± 01.12	96.17 ± 2.03	95.84 ± 2.28	89.74 ± 08.00	98.24 ± 01.31	80.16 ± 5.06	**98.88 ± 1.02**
Banana (10%)	56.18 ± 12.43	55.62 ± 13.25	89.06 ± 12.97	84.38 ± 08.83	90.19 ± 3.85	86.79 ± 2.15	89.44 ± 05.25	72.59 ± 13.55	56.18 ± 12.43	**90.95 ± 4.42**
Bupa	60.95 ± 9.04	65.00 ± 7.888	49.63 ± 5.01	67.05 ± 09.62	**68.63 ± 5.28**	57.79 ± 6.62	57.16 ± 08.84	58.77 ± 11.71	64.95 ± 9.04	63.92 ± 6.85
Cleveland	55.14 ± 6.92	48.83 ± 06.64	55.54 ± 2.76	54.55 ± 05.65	56.92 ± 6.92	50.23 ± 6.76	**56.99 ± 06.61**	55.80 ± 8.29	55.13 ± 6.92	51.24 ± 8.49
Glass	68.13 ± 10.26	**68.95 ± 10.78**	51.11 ± 17.89	66.71 ± 12.16	68.30 ± 11.82	67.02 ± 10.40	60.66 ± 08.20	64.65 ± 9.33	68.13 ± 10.25	66.94 ± 13.24
Ionosphere	85.18 ± 4.92	84.04 ± 03.43	86.16 ± 19.30	**94.02 ± 02.37**	92.89 ± 2.89	88.88 ± 4.15	86.33 ± 07.32	92.60 ± 3.39	85.18 ± 4.92	89.47 ± 4.59
Iris	94.00 ± 4.67	96.00 ± 04.00	**96.67 ± 4.47**	**96.67 ± 04.47**	96.00 ± 2.72	95.33 ± 4.27	94.67 ± 04.00	**96.67 ± 3.33**	94.00 ± 4.67	96.00 ± 3.26
Letter (10%)	78.93 ± 10.95	**85.7 ± 11.83**	25.95 ± 21.15	77.64 ± 08.40	46.54 ± 23.79	61.55 ± 21.08	46.10 ± 08.06	46.55 ± 23.79	79.13 ± 8.95	77.77 ± 4.72
Magic (10%)	62.28 ± 9.68	77.34 ± 09.54	73.5 ± 5.57	**84.65 ± 02.74**	84.07 ± 2.64	67.99 ± 9.85	82.07 ± 02.58	84.57 ± 2.60	62.82 ± 9.68	83.49 ± 2.58
Monk-2	95.89 ± 3.92	**100 ± 0.00**	71.44 ± 12.13	96.52 ± 03.29	98.37 ± 1.48	**100 ± 0.00**	95.21 ± 04.92	98.86 ± 1.52	98.18 ± 2.72	**100 ± 0.00**
Movement-libras	80.83 ± 3.39	87.22 ± 05.28	53.33 ± 21.64	82.78 ± 04.44	75.00 ± 10.54	81.67 ± 4.51	33.61 ± 11.81	67.78 ± 20.38	80.12 ± 3.39	**87.78 ± 3.61**
Newthyroid	95.36 ± 2.95	94.87 ± 03.58	97.25 ± 3.16	96.77 ± 03.57	95.84 ± 3.84	**97.73 ± 3.05**	96.32 ± 04.95	97.25 ± 3.65	95.37 ± 2.95	96.77 ± 2.11
Phoneme (10%)	81.91 ± 8.87	81.91 ± 08.70	70.16 ± 9.79	82.39 ± 04.97	79.96 ± 5.88	**83.4 ± 9.81**	75.73 ± 04.41	76.64 ± 8.12	62.71 ± 8.87	81.48 ± 4.31
Pima	72.93 ± 4.30	69.28 ± 04.04	64.46 ± 4.65	74.01 ± 05.96	**74.11 ± 3.96**	70.44 ± 3.78	72.80 ± 05.22	72.55 ± 3.94	72.93 ± 4.30	70.44 ± 3.52
Ring (10%)	51.70 ± 6.39	59.61 ± 07.46	73.15 ± 11.33	79.45 ± 19.21	**93.91 ± 4.84**	51.19 ± 9.57	82.71 ± 04.11	70.10 ± 20.91	51.77 ± 6.39	89.22 ± 4.56
Satimage (10%)	84.18 ± 9.65	74.58 ± 09.15	59.50 ± 17.88	**86.35 ± 03.12**	82.89 ± 4.25	64.93 ± 21.39	69.82 ± 11.49	83.99 ± 4.72	52.93 ± 19.66	84.11 ± 3.57
Segment (10%)	73.12 ± 12.48	83.69 ± 04.05	67.11 ± 17.72	84.76 ± 07.22	92.21 ± 4.68	64.88 ± 15.37	57.02 ± 16.15	**92.74 ± 3.85**	53.13 ± 22.48	89.76 ± 3.96
Sonar	83.07 ± 10.58	82.09 ± 12.12	59.26 ± 10.03	86.00 ± 10.92	**88.43 ± 9.13**	86.50 ± 9.55	79.33 ± 11.93	85.05 ± 10.91	82.70 ± 10.57	83.59 ± 11.26
Thyroid (10%)	90.99 ± 2.07	92.65 ± 02.21	91.68 ± 1.79	94.48 ± 02.15	93.69 ± 2.19	90.14 ± 2.18	93.74 ± 02.50	64.17 ± 31.81	90.99 ± 2.07	**94.71 ± 2.45**
Titanic (10%)	61.74 ± 16.72	71.82 ± 12.67	73.13 ± 5.05	72.67 ± 05.23	73.18 ± 5.16	64.16 ± 9.05	72.67 ± 05.34	73.61 ± 5.78	**74.14 ± 10.96**	66.97 ± 15.85
Twonorm (10%)	93.83 ± 4.54	95.95 ± 04.39	91.89 ± 10.69	95.95 ± 06.07	96.62 ± 1.38	95.95 ± 8.16	**98.11 ± 01.38**	93.78 ± 10.22	97.5 ± 4.59	97.43 ± 1.76
Vehicle	70.21 ± 3.66	65.95 ± 03.46	40.23 ± 10.55	74.13 ± 12.02	81.91 ± 4.02	74.71 ± 3.13	45.17 ± 08.40	**85.58 ± 3.45**	65.95 ± 3.66	84.40 ± 2.63
Vowel	97.77 ± 0.98	98.28 ± 01.37	88.69 ± 5.58	57.37 ± 15.63	95.25 ± 3.10	97.68 ± 1.20	72.12 ± 06.70	78.79 ± 10.75	98.28 ± 0.98	**98.99 ± 0.90**
Wdbc	96.48 ± 2.49	92.79 ± 03.32	95.61 ± 4.72	**97.36 ± 02.40**	95.18 ± 1.97	92.44 ± 3.16	93.86 ± 07.49	96.48 ±2.72	92.79 ± 2.49	96.65 ± 2.89
Wine	95.52 ± 4.17	97.71 ± 02.80	95.51 ± 4.72	97.22 ± 05.12	82.56 ± 2.22	97.44 ± 2.54	89.25 ± 13.83	96.63 ± 5.11	69.18 ± 4.16	**97.77 ± 5.11**
Wisconsin	96.52 ± 2.79	**96.66 ± 02.71**	74.78 ± 2.51	96.05 ± 01.84	96.05 ± 2.36	95.82 ± 1.73	95.92 ± 02.58	96.22 ± 2.33	96.39 ± 2.94	96.21 ± 2.26
Accuracy_AVG	0.7768	0.7827	0.6923	0.8076	0.8115	0.7732	0.7380	0.7801	0.7486	**0.8268**
Ranking_AVG	5.71429	4.96429	7.17857	3.7857	3.9643	5.67857	6.2143	4.7857	5.89286	**4.3214**
Diff_AVG	0.0726	0.0667	0.1571	0.0418	0.0379	0.0762	0.1114	0.0693	0.1007	**0.0242**
# of 1st	0	5	1	5	4	3	2	3	1	**7**

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